

How to Read a Safety Data Sheet (SDS)

Hoke Industrial, LLC | Hazard Communication (OSHA 29 CFR 1910.1200)

An SDS is the manufacturer's safety manual for a product. Before you use, store, or clean up a chemical, the SDS tells you the hazards, the protective gear, and exactly what to do in an emergency. Every SDS uses the same 16 sections in the same order — here's how to read one fast.

1. Know the warning pictograms (found in Section 2 of every SDS)



Flammable

Catches fire easily — liquids, gases, aerosols, solids.



Corrosive

Burns skin & eyes; can eat through metal.



Harmful / Irritant

Irritation, dizziness; less-serious health effects.



Health Hazard

Carcinogen, lung/organ damage, sensitizer, reproductive.



Toxic

Poison — can be fatal even in small amounts.



Oxidizer

Feeds a fire / makes it burn hotter.



Gas Under Pressure

Compressed gas; cylinder can burst or explode.



Explosive

May explode from heat, shock, or flame.



Environmental

Toxic to fish & aquatic life — keep out of drains.

2. Check the signal word (Section 2)

DANGER	More severe hazard. Take maximum precautions.
WARNING	Less severe hazard — still requires care and proper PPE.

3. In an emergency, jump straight to these sections

Someone exposed / hurt	Section 4 — First-Aid Measures
Fire	Section 5 — Fire-Fighting Measures
Spill or leak	Section 6 — Accidental Release
What PPE to wear	Section 8 — Exposure Controls / PPE

Page 2 explains all 16 sections in order.

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The 16 sections of every SDS

U.S. law (OSHA HazCom 2012 / GHS) requires every SDS to use these 16 sections in this exact order. The “>>” mark flags the sections you’ll reach for most. Sections 12–15 are not enforced by OSHA but are normally included.

#	Section	What it tells you
1	Identification	Product name, maker, emergency phone #, recommended use.
2	Hazard(s) Identification	The hazards: pictograms, signal word (Danger/Warning), hazard statements. Read this first.
3	Composition / Ingredients	What’s in it — hazardous chemicals, CAS #s, amounts.
4	First-Aid Measures	>> What to do if exposed — eyes, skin, breathing, swallowing.
5	Fire-Fighting Measures	>> Right extinguisher to use; fire/explosion hazards.
6	Accidental Release (Spill)	>> How to contain & clean up a spill or leak safely.
7	Handling & Storage	Safe handling, storage conditions, what to keep it away from.
8	Exposure Controls / PPE	>> Exposure limits + the gloves, eyewear, respirator to wear.
9	Physical & Chemical Properties	Looks, smell, flash point, boiling point, pH, etc.
10	Stability & Reactivity	Is it stable? What makes it react; what it’s incompatible with.
11	Toxicological Information	Health effects, symptoms, routes of exposure, cancer info.
12	Ecological Information	Environmental effects (not OSHA-enforced).
13	Disposal Considerations	How to dispose of it properly (not OSHA-enforced).
14	Transport Information	DOT/shipping info — UN number, class (not OSHA-enforced).
15	Regulatory Information	Other safety/health/environmental regulations (not OSHA-enforced).
16	Other Information	Revision date, what changed, legend/abbreviations.

Bottom line: open to Section 2 to size up the hazard, grab your PPE from Section 8, and in an emergency go to Sections 4, 5, or 6. If you can’t find the SDS, ask a supervisor — never use a product whose SDS you can’t locate.